

Task 1: GDPR vs HIPAA vs PCI DSS

Standard	Key Cloud Security Requirements
GDPR	Protect personal data; implement appropriate security measures; control access to personal information and report certain data breaches.
HIPAA	Protect electronic protected health information (ePHI); implement access controls; maintain audit controls and security safeguards.
PCI DSS	Protect cardholder data; restrict access based on business need; monitor and test networks; maintain secure systems and authentication mechanisms.

Task 2: AWS Trusted Advisor

Steps

1. Log in to AWS.
2. Open Trusted Advisor.
3. Review the available checks.
4. Look for security-related checks such as:
 - MFA on root account
 - Security Groups
 - S3 permissions
5. Run or refresh available checks.

Task 3: Analyze a Security Issue

Example: Missing Root MFA

Security issue: MFA is not enabled on the AWS root account.

Risk

The AWS root account has highly privileged access. If its password is compromised, an attacker could potentially make significant changes to cloud resources.

Remediation

Enable multi-factor authentication (MFA) on the AWS root account and use strong, unique credentials.

Compliance Connection

Failure to properly protect privileged access could contribute to non-compliance with security requirements under standards such as GDPR, HIPAA or PCI DSS, depending on the data and environment involved.

Task 4: Enable Root MFA

Steps

1. Sign in as the AWS root user.
2. Open the account/security credentials area.
3. Locate Multi-Factor Authentication (MFA).
4. Select Assign MFA device.
5. Choose an appropriate MFA method.
6. Follow the enrollment process.
7. Verify the MFA device.
8. Sign out and test the MFA login.

Task 5: Before-and-After Trusted Advisor

Before

Example:

Check	Before
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Root MFA	Warning
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Security Groups	Warning
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S3 Permissions	OK
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Action Taken

Enabled MFA on the root account.

Check	After
Root MFA	OK
Security Groups	Warning
S3 Permissions	OK